

Cantilever characterisation using Position Sensitive detector

Final Project Report Submission

Project Members:

Anveesha Arora (ID: 5464416)
Donna Alex (ID: 5464375)
Georgey Baby Valavathu (ID: 5462438)
Nithish Kumar Baskaran (ID: 5462454)
Nigill Jayaraman Kavitha (ID: 5462784)
Nandini Mavillapalle (ID: 5464247)
Nikhila Nallam (ID: 5464391)

Professors:

Prof. Dr. Ing. Friedrich Fleischmann
Prof. Dr. rer. nat. Ludger Kempen

Declaration of independent preparation of work

I hereby declare that I have written the present work independently and have not used any sources or aids other than those indicated.

The passages of this work that were taken from other works, either verbatim or in terms of meaning, are marked with indications of their origin.

This declaration also applies to the graphics, sketches, and visual representations contained in the work. I have not submitted this work, either in the same or similar form, nor in excerpts, as part of an examination or academic requirement.

I certify that the electronic version of the submitted work is identical to the printed version, and I agree to an electronic review of the work using plagiarism detection software.

Please select one of the following options; one of the two options must be chosen in agreement between the examiner and the examinee:

Option 1: Use of AI-based Tools without Disclosure Requirement

I certify that I have not used any AI-based tools whose use has not been agreed upon in writing with the examiner. I am aware that the use of texts or other content and products generated by AI-based tools does not guarantee their quality. I assume responsibility for the work submitted and for its content. I further certify that my own creative input predominates in the present work.

✓ Option 2: Use of AI-based Tools with Disclosure Requirement

I certify that I have not used any AI-based tools whose use has not been agreed upon in writing with the examiner. I am aware that the use of texts or other content and products generated by AI-based tools does not guarantee their quality. I assume responsibility for the work submitted and for its content. I further certify that my own creative input predominates in the present work. All verbatim or paraphrased contributions and quotations, as well as all sections created, written, and/or edited with the aid of AI-based tools, are marked and documented by me. The form of disclosure will be agreed upon between the examiner and the examinee.

I am aware that a violation of the points mentioned may have examination-related consequences and, in particular, may lead to the examination performance being graded as “insufficient” or the academic performance as “failed.” In cases of repeated or serious misconduct, expulsion may occur.

	First name and surname	Student ID Number
1.	Anveesha Arora	5464416
2.	Donna Alex	5464375
3.	Georgey Baby Valavathu	5462438
4.	Nithish Kumar Baskaran	5462454
5.	Nigill Jayaraman Kavitha	5462784
6.	Nandini Mavillapalle	5464247
7.	Nikhila Nallam	5464391

Contents

1	Introduction	4
2	Objective of the Project	4
3	Expected Outcomes	5
3.1	Deliverables	5
3.2	Quantitative Results	5
3.3	Validation Outcome	6
4	Experimental Setup	6
4.1	Physical	6
4.2	Position Sensitive detector	6
4.3	Hypothesis and mathematical calculations	7
4.3.1	Law of reflection	7
4.3.2	Natural frequency	8
4.3.3	Damping constant	9
4.3.4	Quality Factor	10
5	Amplifier Design	11
5.1	Transimpedance Amplifier Design Considerations	11
5.1.1	Current-to-Voltage Conversion	11
5.1.2	Component Selection: The MCP6002 Dual Op-Amp	11
5.1.3	The Stability Crisis: Phase Margin and Cf Compensation	11
5.1.4	The R_f and Bandwidth Trade-off	12
5.2	SPICE Modeling and Parameter Optimization	12
5.2.1	S3932 Equivalent Circuit in LTspice	12
5.2.2	Inference from SPICE Simulation	13
5.3	Differential Signal Extraction	13
5.3.1	Transition to Voltage-Stage Processing	13
5.3.2	Implementation with the LM324 Quad Op-Amp	13
6	Results	15
6.1	Data analysis techniques	15
6.1.1	Fourier Transform	15
6.1.2	Peak detection method	16
6.1.3	Zero-crossing method	18
6.1.4	Parabolic interpolation of the FFT peak	19
6.2	Comparison between different cantilevers	21
7	Conclusion	21
8	Contributions from team members	22
9	Documentation of Tools and AI Usage	23
9.1	Generative AI Usage	23
9.2	Software and Hardware Tools	23
	References	24

List of Figures

1	Overall signal chain, from the optical path (laser – mirrors – cantilever – PSD) to the electrical path (TIA – differential amplifier – PicoScope).	5
2	Schematic of a PSD cross-section. [6]	7
3	LT SPICE testbench for the dual-Transimpedance Amplifier and S3932 equivalent circuit.	12
4	Transient Analysis of the Testbench	13
5	KiCad schematic of the final amplifier hardware setup	14
6	Flowchart for the Fourier Transform method.	15
7	FFT spectra of the three cantilever configurations, showing the shift in dominant frequency between the Normal (474.0 Hz), Mass (290.7 Hz) and Drop (358.3 Hz) cases.	16
8	Flowchart for the peak detection method.	17
9	Peak detection and exponential decay fit for the three cantilever configurations.	17
10	Flowchart for the zero-crossing frequency calculation.	18
11	Zero-crossing method applied to the three cantilever configurations.	19
12	Flowchart for the parabolic interpolation method.	20
13	Parabolic interpolation of the FFT peak for the three cantilever configurations, showing the discrete FFT bins, the fitted log-parabolic curve, the standard discrete peak, and the refined sub-bin peak.	20

List of Tables

1	Effect of Beam Parameters on Natural Frequency	8
2	Eigenvalue Constants for Different Modes (Fixed-Free Cantilever)	8
3	Effect of Damping Parameters on Cantilever Vibration	10
4	Standard discrete FFT peak vs. parabolically refined peak.	21
5	Comparison of the three cantilever configurations.	21
6	Documentation of Generative AI Usage	23
7	Documentation of Engineering Tools and Software	23

1 Introduction

Cantilever-type resonant structures are often used as transducers for force, mass, and vibration measurements. Such structures find application in atomic force microscope (AFM) probes, MEMS resonators, and acoustic sensors. Two parameters define the dynamics of a cantilever structure: its natural frequency, which determines the measurement bandwidth and sensitivity, and its damping, which in turn defines the decay rate of the free oscillations and hence the system's frequency selectivity.

A common way to observe the motion of a cantilever is the optical lever technique, which uses a laser spot reflected from the cantilever and directed at a position-sensitive photodetector. As the laser beam path is much longer than the cantilever itself, tiny rotations of the latter are amplified and turned into significant movements of the spot on the photodetector [8]. The optical lever technique is the standard method for measuring the position of a cantilever in AFMs and similar structures due to its extraordinary sensitivity (resolution of the order of 10^{-3} nm). It is also relatively easy to implement compared to other available methods, notably interferometry. [8],[4].

This project measures the natural frequency and damping of a cantilever structure by observing its oscillations with an optical lever. The position of the laser spot is picked up by a Position Sensitive Detector (PSD) and converted into a voltage via an analog circuitry. The voltage signal is digitized and processed in the time domain with Python in a PicoScope oscilloscope. A similar approach is used to process the signal in the frequency domain. The quality factor derived from the analysis is of particular interest to the project, as it describes the resonance characteristics of the cantilever and determines the rate at which its oscillations will decay when undriven.[1].

2 Objective of the Project

The main goal of the project is to design an optical-lever measurement chain, which could capture the free-vibration response of a cantilever. A signal-processing procedure was also designed to analyze the response function and identify the natural frequency, damping ratio, and quality factor. Particular goals of this project include:

1. Set up an optical lever arrangement in such a way that a laser beam is reflected off of the vibrating cantilever using a folding mirror and then is reflected off a second mirror onto a Position Sensitive Detector.
2. The PSD output has to be conditioned by means of the transimpedance amplifier (TIA) followed by the differential amplifier resulting in a single output voltage proportional to the angular position of the cantilever.
3. Capture the conditioned analog signal in a digital format using PicoScope data-acquisition software at a sampling rate adequate to identify the cantilever's resonant frequency and its decay envelope.
4. Process the received signal in Python to find the resonant frequency, and to estimate the damping ratio and quality factor of the cantilever using both time (ring-down) and frequency (spectral) domain methods.
5. Cross-validate the frequency and damping estimates determined by the two independent analysis methods and document the measurement chain, procedure and results in a technical report.

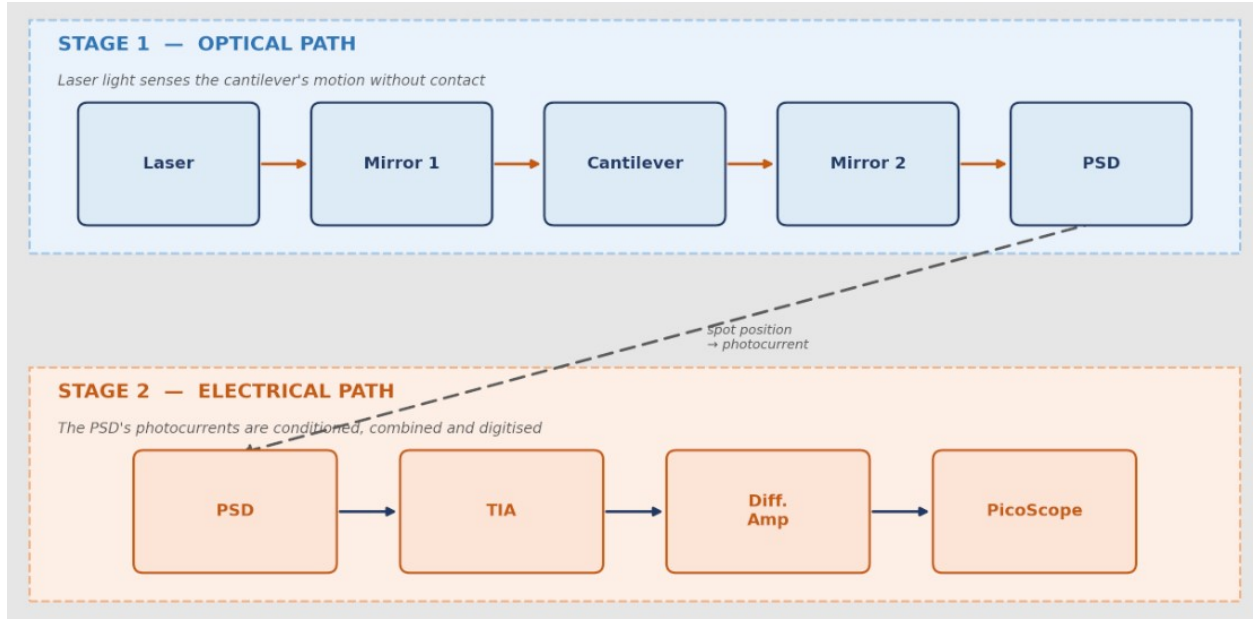


Figure 1: Overall signal chain, from the optical path (laser – mirrors – cantilever – PSD) to the electrical path (TIA – differential amplifier – PicoScope). .

3 Expected Outcomes

The protocol the considerations are expected to produce a working measurement chain, an analysis pipeline that actually extracts useful parameters from the measurement, and produce quantified results that make physical sense on the cantilever under test. With this in mind, the following sections outline what are considered to be the deliverables, quantitative results, validation outcome of the project.

3.1 Deliverables

A validated optical-lever measurement chain, from the laser and mirrors through the PSD, transimpedance amplifier and differential amplifier to PicoScope digitisation, that can reliably reproduce a clean ring-down trace without being clipped at any point during repeated measurements.

A Python analysis pipeline that can reliably extract the cantilever's natural frequency, damping ratio and quality factor from the ring-down signal, separately implementing time- and frequency-domain fitting procedures as individual, reusable functions.

3.2 Quantitative Results

The analysis pipeline's key result is a self-consistent set of natural frequency (f_0), damping ratio (ζ) and quality factor (Q) for the cantilever under test, each derived twice (once from the time domain trace and once from the frequency domain spectrum), For a lightly damped system such as a cantilever this implies that the ring-down trace will exhibit a well-defined oscillation frequency with an exponentially decaying amplitude envelope, and likewise the spectrum will feature a single narrow peak at the natural frequency f_0 . A physically reasonable damping ratio for such a system would be in the range $\zeta = 10^{-3}$ to 10^{-2} , corresponding to a quality factor of approximately few tens to few hundreds for macro-scale cantilevers vibrating in air [5][1]. The project's quantitative results are expected to take the form of the actual measured values of f_0 , ζ , and Q for the cantilever under test, along with an indication of the magnitude of the discrepancy (in percentage terms) between the two independent measurements as a confirmation of consistency.

3.3 Validation Outcome

A successful outcome for the project would be a demonstration that the independently measured natural frequencies and quality factors from the time- and frequency-domain measurements are consistent with each other (ideally within a few percent). This would demonstrate that the combination of the optical lever measurement chain and the analysis pipeline is capable of reliably capturing the dynamics of a lightly damped cantilever system. A larger discrepancy between the two would also be informative, as it would identify a particular link in the measurement chain (most likely the amplifier bandwidth or the frequency resolution of the Picoscope) that failed to meet the requirements of the project.

4 Experimental Setup

4.1 Physical

The physical apparatus consisted of the following components:

- **Mounting Hardware:** The optical components were securely mounted using optical posts, kinematic mirror mounts, and a fixed clamp to hold the cantilever wafer. All components were installed on a vibration-damped optical breadboard to reduce unwanted mechanical vibrations and improve measurement stability.
- **Laser diode module:** The laser diode module has a built-in, adjustable internal focusing lens. The position of the lens is adjusted by rotating the barrel of the module, allowing the beam to be focused to a small spot on the cantilever surface. The laser diode was powered by a 5 V power supply.
- **Steering mirrors (M1, M2):** Two front-surface mirrors mounted on adjustable kinematic mounts, used to route the beam from the laser onto the cantilever (M1) and from the cantilever onto the detector (M2).
- **Microcantilever:** Clamped at its fixed end within a chip package. The free end carries a reflective coating that acts as the reflecting element for the optical lever. Three different cantilevers were compared to study the effect of added mass:
 - (i) a bare cantilever (no added mass), used as a reference;
 - (ii) a cantilever with a drop of glue at the free end;
 - (iii) a cantilever loaded with an additional mass.
- **Position-Sensitive Detector (PSD):** A Hamamatsu S3932 one-dimensional lateral-effect photodiode was used to detect the laser beam position. The device has two anode terminals (Anode X_1 and Anode X_2), and the relative photocurrents from these terminals indicate the position of the incident beam spot along the sensing axis.
- **Transimpedance Amplifiers (TIAs):** Two matched transimpedance amplifier (TIA) circuits were built using MCP6002 opamps and were used to convert the photocurrent from each anode into a voltage signal. Each TIA employs a feedback resistor of $R_f = 100 \text{ k}\Omega$ to perform the current-to-voltage conversion.
- **Instrumentation Amplifier:** The instrumentation amplifier built using LM324n opamp. The voltage outputs V_a and V_b from the two transimpedance amplifiers were fed into the input stage, where a $7 \text{ k}\Omega$ gain resistor (R_g) determined the amplification. The last op-amp compared the two signals and gave a single differential output, $V_{\text{out}} = V_a - V_b$, which was the voltage difference between the two inputs.
- **Data Acquisition Device:** A PicoScope 2000 series USB oscilloscope was used to record $V_{\text{out}}(t)$.
- **Mechanical Vibration:** To vibrate the cantilever, the mounting clamp holding the cantilever chip in place was lightly tapped. This was enough to set the cantilever oscillating freely at its natural frequency.

4.2 Position Sensitive detector

A PSD (Position Sensitive detector) is an optical sensor consisting of a uniform resistive layer, that is able to translate the position of an incident light beam to current signals. The size and fast response of the PSD makes it able to measure microscopic movements in real time with precision. A PSD in general is a semiconductor device consisting of -

- A uniform resistive layer
- A P-type layer
- Intrinsic layer

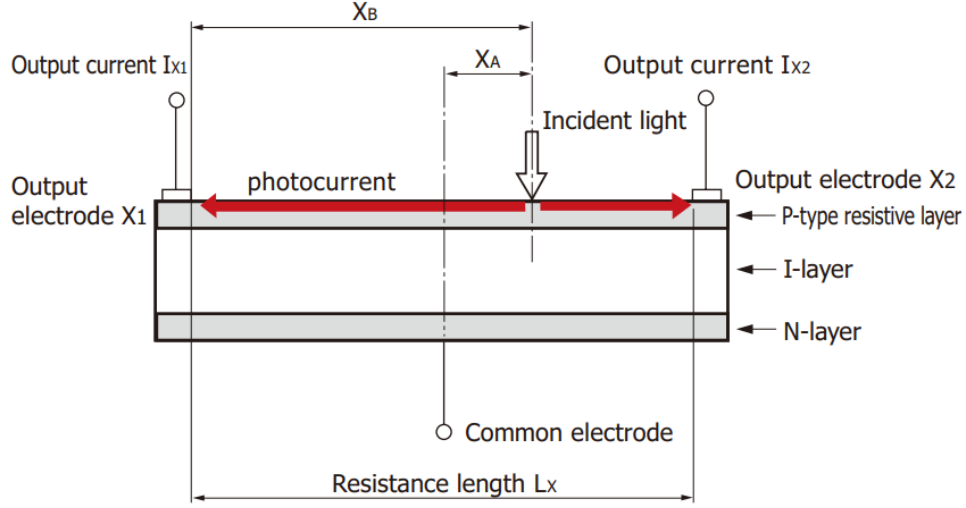


Figure 2: Schematic of a PSD cross-section. [6]

- N-type layer
- two output and one common electrode.

When a light beam is incident on the surface of the resistive layer of the PSD, a current is generated relative to the position of the incident beam. The photocurrent generated reaches across the electrodes which is also divided proportionally to the distance between the electrodes and the incident beam. The position of the incident beam can be calculated by the formula below -

$$x = \frac{l_x}{2} \cdot \frac{I_{x1} - I_{x2}}{I_{x1} + I_{x2}} \quad (1)$$

where:

l_x is the length of the resistive layer (mm),

x is the position of the incident beam,

I_{x1} is the current across first electrode (usually in mA).

I_{x2} is the current across second electrode.

4.3 Hypothesis and mathematical calculations

4.3.1 Law of reflection

The laser displacement measurement used in this project is based on the **law of reflection**. According to this law, when a beam of light strikes a smooth reflective surface, the angle of incidence is equal to the angle of reflection.

$$\theta_i = \theta_r \quad (2)$$

where:

θ_i is the angle of incidence,

θ_r is the angle of reflection.

Both angles are measured with respect to the normal, which is an imaginary line perpendicular to the reflecting surface. In this experiment, a laser beam is directed onto a mirror attached to the free end of the cantilever beam. As the cantilever vibrates, the mirror rotates through a small angle. According to the law of reflection, the reflected laser beam changes its direction, allowing the beam deflection to be measured accurately. For small angular displacements, the reflected beam rotates through twice the angle of the mirror rotation. Therefore,

$$\theta_r = 2\theta \quad (3)$$

where:

θ is the angular displacement of the mirror,
 2θ is the angular displacement of the reflected laser beam.

If the screen is placed at a distance D from the mirror, the displacement of the reflected laser spot is approximately

$$x \approx 2D\theta \quad (4)$$

where:

x is the displacement of the laser spot (m),
 D is the distance between the mirror and the screen (m),
 θ is the angular displacement of the mirror (rad).

This relationship enables accurate measurement of very small vibrations of the cantilever beam.

4.3.2 Natural frequency

The cantilever beam is a structural element which is rigidly fixed at one end but free at the other end. The boundary condition plays a vital role in determining the dynamic characteristics of the cantilever beam. When a cantilever beam is subjected to an external disturbance, it vibrates at specific characteristic frequencies called natural frequencies, which are dependent on the stiffness and mass distribution of the beam. The natural frequency for the fundamental (first) mode of vibration of a uniform cantilever beam is obtained from the Euler-Bernoulli beam theory and is expressed as:

$$f_1 = \frac{1}{2\pi} \beta_1^2 \sqrt{\frac{EI}{mL^4}} \quad (5)$$

where:

f_1 is the natural frequency (in Hz),
 E is the modulus of elasticity of beam material (in N/m²),
 I is the area moment of inertia of the beam cross-section (in m⁴),
 m is the mass per unit length of the beam (in kg/m),
 L is the length of the beam (in m),
 β_1 is the eigenvalue constant for the first mode of a fixed-free beam ($= 1.875$).

The effect of these parameters on the natural frequency is summarized in Table 1.

Table 1: Effect of Beam Parameters on Natural Frequency

Parameter	Effect on Natural Frequency	Reason
Increase in E or I (stiffness $\uparrow$)	Frequency increases	Higher stiffness resists deformation
Increase in L (length $\uparrow$)	Frequency decreases sharply	L^4 term in denominator dominates
Increase in m (mass $\uparrow$)	Frequency decreases	Higher inertia slows vibration

With higher modes, the eigenvalue constant increases as well, resulting in higher natural frequencies and more intricate mode shapes, as illustrated in Table 2.

Table 2: Eigenvalue Constants for Different Modes (Fixed-Free Cantilever)

Mode Number	Eigenvalue Constant (β_n)	Description
1	1.875	Fundamental mode
2	4.694	Second mode
3	7.855	Third mode
4	10.996	Fourth mode

4.3.3 Damping constant

When a cantilever beam is set into vibration, its amplitude gradually decreases with time because some of its mechanical energy is lost to the surroundings. This loss of energy is mainly caused by air resistance, internal friction within the material and other damping effects. This phenomenon is known as **damping**. The damping constant describes how quickly the vibration decays. A larger damping constant causes the oscillations to die out more rapidly, whereas a smaller damping constant allows the beam to continue vibrating for a longer period.

The displacement of a damped cantilever beam is expressed as

$$x(t) = Ae^{-\delta t} \cos(\omega_d t + \phi) \quad (6)$$

where:

$x(t)$ is the displacement of the beam at time t ,

A is the initial amplitude,

δ is the damping constant (s^{-1}),

ω_d is the damped natural angular frequency (rad/s),

ϕ is the initial phase angle,

t is the time (s).

One of the most commonly used methods to determine the damping constant is the logarithmic decrement method, which compares the amplitudes of successive vibration peaks. The logarithmic decrement is given by

$$\Lambda = \ln \left(\frac{x_1}{x_2} \right) \quad (7)$$

where:

x_1 is the first peak amplitude,

x_2 is the second peak amplitude.

If the amplitudes are measured over n vibration cycles, the logarithmic decrement becomes

$$\Lambda = \frac{1}{n} \ln \left(\frac{x_1}{x_{n+1}} \right) \quad (8)$$

The damping ratio is calculated using

$$\zeta = \frac{\Lambda}{\sqrt{4\pi^2 + \Lambda^2}} \quad (9)$$

The damping constant is then obtained from

$$\delta = \zeta \omega_n \quad (10)$$

where:

ζ is the damping ratio,

ω_n is the natural angular frequency (rad/s).

The natural angular frequency is related to the natural frequency by

$$\omega_n = 2\pi f_n \quad (11)$$

where:

f_n is the natural frequency (Hz).

The effect of the damping parameters on the vibration response is summarized in Table 3.

Table 3: Effect of Damping Parameters on Cantilever Vibration

Parameter	Effect on Vibration	Reason
Increase in δ	Faster amplitude decay	Greater energy dissipation
Decrease in δ	Slower amplitude decay	Lower energy dissipation
Increase in ζ	Reduced oscillations	Higher damping suppresses vibration
Increase in ω_n	Faster oscillation	Higher natural frequency of the system

4.3.4 Quality Factor

The quality factor Q describes how sharp the resonance of the cantilever is and how slowly its vibration dies away after it is excited. It is a dimensionless number that compares the energy stored in the oscillation to the energy lost in each cycle. A high Q means the cantilever keeps vibrating for a long time with very little energy loss, while a low Q means the vibration decays quickly. The quality factor is calculated directly from the damping ratio using the relation

$$Q = \frac{1}{2\zeta}. \quad (12)$$

A larger Q therefore corresponds to a smaller damping ratio.

5 Amplifier Design

The resonant frequency and damping ratio are the major functions to determine characterization of a micro cantilever. The method of optical lever uses a Position Sensitive Detector (Hamamatsu S3932) to convert the physical deflection into an electrical signal. The S3932, transduces a small photocurrent as a result of physical deflection of a micro cantilever. Additionally, the inherent capacitance of the Position Sensitive Detector interacts with the amplifier circuitry to cause severe instability at high frequencies. So, In order to analyze a meaningful signal change from the photo detector a well tuned amplifier circuitry is needed. This chapter describes the design consideration, SPICE simulation and the hardware implementation of the transimpedance and differential amplifier stages required to convert the Position Sensitive Detector current into a clean and suitable signal for processing.

5.1 Transimpedance Amplifier Design Considerations

5.1.1 Current-to-Voltage Conversion

A Position Sensitive Detector (PSD) basically consists of a uniform resistive layer formed on one or both surfaces of a high-resistivity semiconductor substrate and a pair of electrodes formed on both ends of the resistive layer for extracting position signals. The photosensitive area, which is also a resistive layer, has a PN junction that generates photocurrent by means of the photovoltaic effect[6]. Therefore, A Transimpedance Amplifier (TIA) is necessary [3]. By connecting the Position Sensitive Detector's anode directly to the op-amp's inverting input, a strict virtual ground is maintained [2].

5.1.2 Component Selection: The MCP6002 Dual Op-Amp

For the Transimpedance Amplifier stage, the MCP6002 dual operational amplifier was selected. The primary justification for this choice is because of it's CMOS architecture which provides an ultra-low input bias current. When dealing with Position Sensitive Detector photocurrents in the nanoamp range, op-amps with higher bias currents would affect the signal, causing severe voltage errors and non-linearity.

5.1.3 The Stability Crisis: Phase Margin and Cf Compensation

The primary challenge in Position Sensitive Detector design is the parasitic junction capacitance of the S3932 ($C_{PSD} \approx 80\text{pF}$) [6]. This capacitance, combined with the feedback resistor (R_f), creates an unwanted pole in the feedback loop [7]. At high frequencies, this pole introduces up to a -90° phase lag. Combined with the op-amp's inherent -180° phase inversion, the total loop phase approaches -270° (equivalent to $+90^\circ$), which allows the circuit to act as an Oscillator. A feedback capacitor (C_f) was placed in parallel with R_f . This capacitor introduces a "zero" into the transfer function, providing a $+90^\circ$ phase lead that counteracts the -90° lag from the PSD. The selection of capacitor value is done through SPICE simulation.

5.1.4 The R_f and Bandwidth Trade-off

The value of R_f decides both the signal gain and the system bandwidth. The Transimpedance Amplifier's bandwidth is approximated by

$$BW = \frac{1}{2 \cdot R_f \cdot C_{PSD}} \quad (13)$$

where:

R_f Feedback Resistor,

C_{PSD} Terminal Capacitance of the Photo Sensitive Detector .

Increasing R_f to increase its gain will reduce the system's bandwidth, which may fail to detect the cantilever's resonant frequency. However, the addition of the compensating capacitor C_f alters this dynamic. C_f forms a broadband noise-gain peaking effect that extends the closed-loop bandwidth significantly beyond the basic RC pole.

5.2 SPICE Modeling and Parameter Optimization

5.2.1 S3932 Equivalent Circuit in LTspice

Figure 3 shows a standard single current source SPICE model is insufficient for high-frequency analysis because it ignores the distributed resistance and the inherent capacitance of the Position Sensitive Detector strip. A 1D Pi-network model was constructed in LT SPICE. Based on the S3932 datasheet (Terminal Capacitance $C_t = 80$ pF, Inter-anode Resistance $R_{AB} = 120$ k Ω), the model utilized:

- Resistors: $R_A = 60$ k Ω , $R_B = 60$ k Ω
- Capacitors: $C_A = 20$ pF, $C_C = 40$ pF, $C_B = 20$ pF
- Power Supply: $V_{DD} = 7$ V, $V_{SS} = -7$ V

To accurately simulate the hardware, the UniversalOpamp2 in LT SPICE was configured with a GBW of 1 MHz to exactly match the frequency characteristics of the chosen MCP6002.

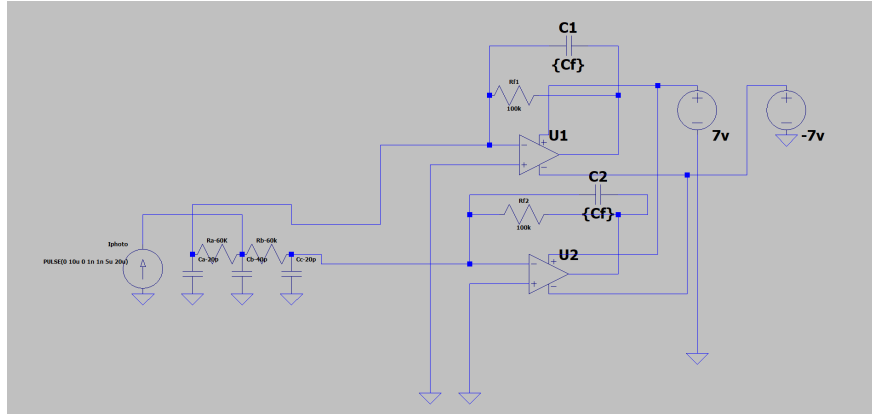


Figure 3: LT SPICE testbench for the dual-Transimpedance Amplifier and S3932 equivalent circuit.

5.2.2 Inference from SPICE Simulation

The Figure 4 shows that a transient response graph is generated with multiple choice of C_f . The SPICE commands for the following analysis are `.tran 0 20u` and `.step param Cf list 0p 10p 20p 30p 40p 50p`. From the analysis we can infer that when the capacitance is zero then there is no high lane for the high frequency signals to escape from the time lag create by the the terminal capacitance C_{PSD} . When the capacitance at that point is very high like 40pF or 50pF, the response becomes noticeably sluggish due to over-damping. The range between 20pF to 30 pF seems very optimal for the application.

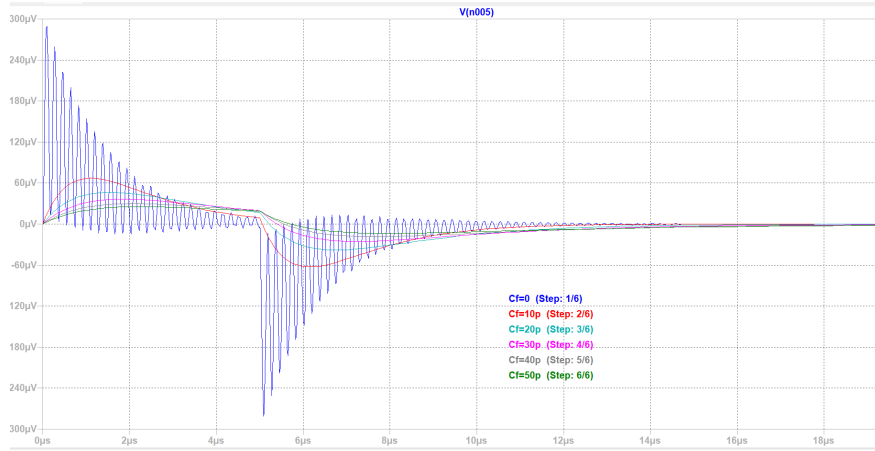


Figure 4: Transient Analysis of the Testbench

5.3 Differential Signal Extraction

5.3.1 Transition to Voltage-Stage Processing

Once the dual MCP6002 TIAs successfully converted the Position Sensitive Detector currents into voltages. For the subsequent mathematical operations—difference calculation ($V_A - V_B$) is needed to find the position of the signal on the Position Sensitive Detector. In order to amplify a difference between 2 signals a differential amplifier is required.

5.3.2 Implementation with the LM324 Quad Op-Amp

The differential stage is implemented using the LM324 which is a quad operational amplifier. While an Instrumentation Amplifier theoretically provides a superior Common-Mode Rejection Ratio and perfectly symmetrical inputs, the LM324 was selected for practical hardware integration. Providing four op-amps in a single package, it minimized board space. In order to have similar input impedance with both inverting and non-inverting input terminal, Both V_A and V_B sent

through a additional non-inverting buffers and with minimal tolerance matching with resistors a differential amplifier circuit can act as an Instrumentation Amplifier with superior Common-Mode Rejection Ratio . To improve the visibility of the signal on the oscilloscope and further optimize the signal-to-noise ratio before digital acquisition, the gain of the differential stage was scaled up.

Figure 5 shows the interconnection between different stages of amplification. Overall power supply of $V_{DD} - V_{EE} = 14V$ is given as power supply to power up the amplifiers.

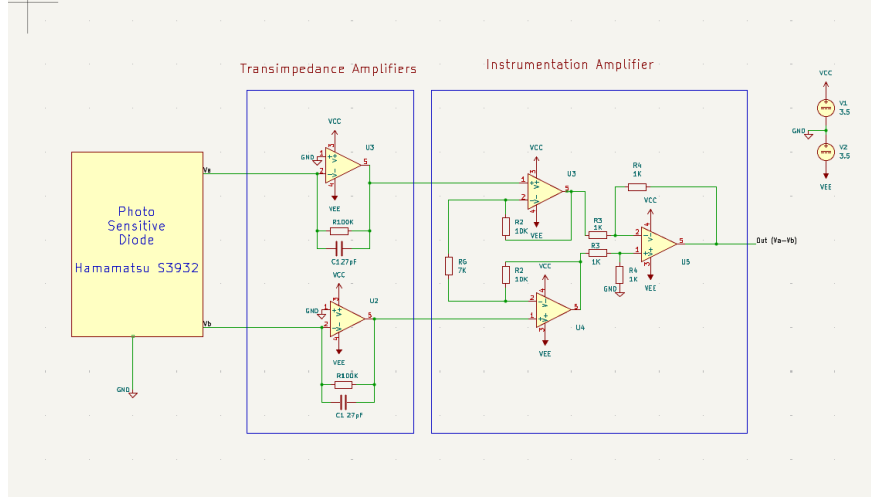


Figure 5: KiCad schematic of the final amplifier hardware setup

The front-end signal conditioning design successfully converted the Hamamatsu S3932 photocurrent into a stable, measurable voltage and with SPICE simulations utilizing equivalent circuit of Position Sensitive Detector selection of R_f and C_f are concluded. The combined use of the Transimpedance Amplifier circuit and Differential Amplifier provided a reliable, practical hardware setup for further signal processing.

6 Results

The signal was processed in Python using Jupyter and was extracted using a Picoscope 2000 series. The section ahead analyses different techniques used to process parameters related to the observed signal. All the source code can be found in the following Github repository.

6.1 Data analysis techniques

6.1.1 Fourier Transform

The Fourier transform is used to change the signal from the time domain to the frequency domain. This shows which frequencies are present in the vibration. It is calculated using the Fast Fourier Transform (FFT), given by

$$X_k = \sum_{n=0}^{N-1} x_n e^{-j2\pi kn/N}. \quad (14)$$

In this method, we usually omit the first few milli seconds of the excitation to avoid processing non linear transients or artifacts. Also for an FFT to accurately measure the natural frequency, the cantilever must be in a state of pure unforced free vibration.

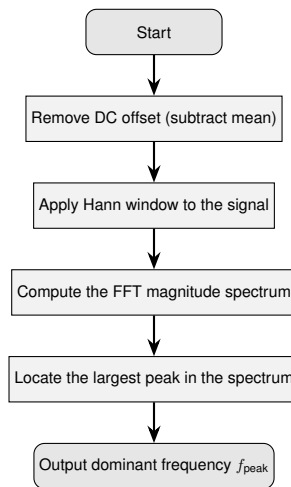


Figure 6: Flowchart for the Fourier Transform method.

FFT spectra of the three cantilevers

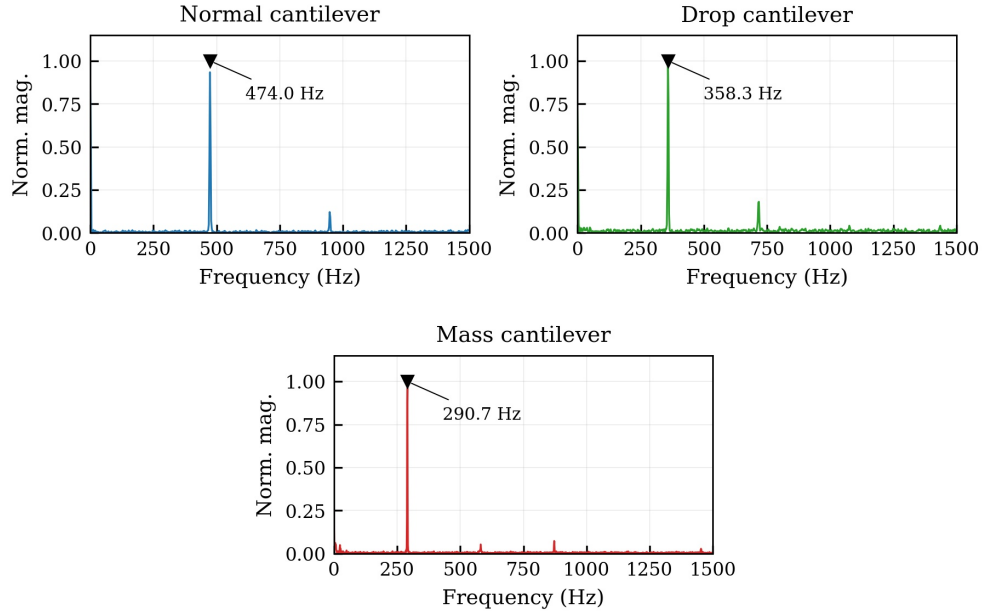


Figure 7: FFT spectra of the three cantilever configurations, showing the shift in dominant frequency between the Normal (474.0 Hz), Mass (290.7 Hz) and Drop (358.3 Hz) cases.

6.1.2 Peak detection method

The peak-detection method works directly with the time-domain signal. It identifies successive maxima in the oscillation, then uses the average time interval between adjacent peaks to determine the vibration period; taking its inverse yields the natural frequency. Since the amplitude decreases exponentially, those same peaks provide an estimate of the damping. The ratio of the first peak amplitude x_0 to a later peak amplitude x_n , separated by n cycles, defines the logarithmic decrement.

$$\delta = \frac{1}{n} \ln\left(\frac{x_0}{x_n}\right) \quad (15)$$

from which the damping ratio is obtained as

$$\zeta = \frac{\delta}{2\pi} \quad (16)$$

For this method, the interval that is chosen to process is the main source of error. The method tries to fit a straight line, using individual data points. Every point that is included or excluded during selection of interval will affect the slope calculated, leading to error in damping ratio.

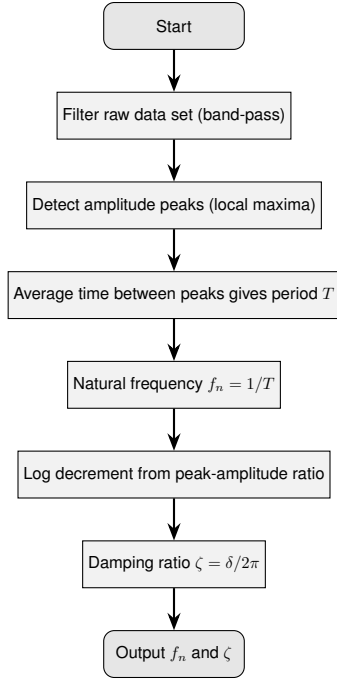


Figure 8: Flowchart for the peak detection method.

Peak detection and exponential decay fit for the three cantilevers

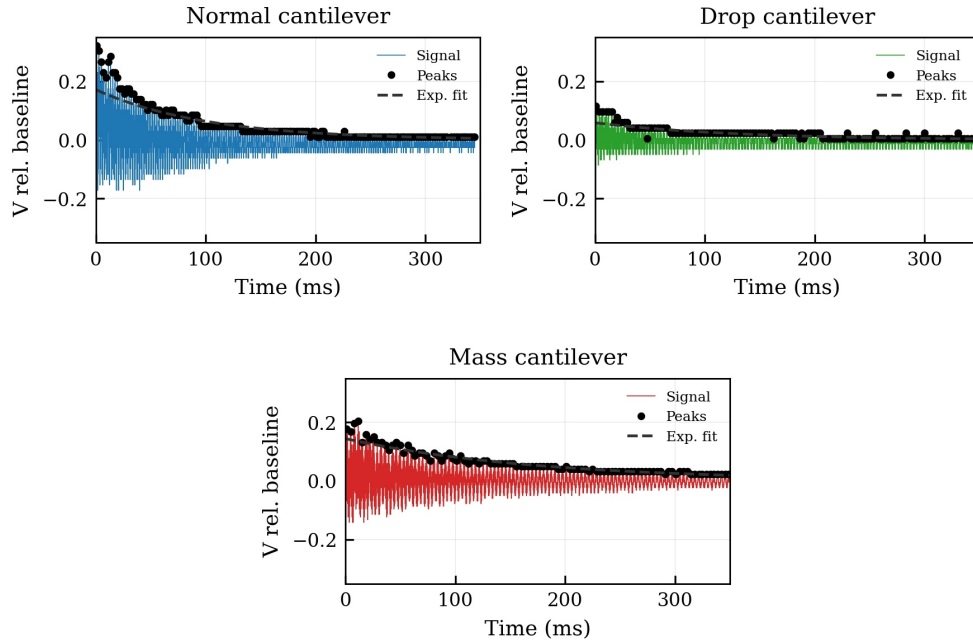


Figure 9: Peak detection and exponential decay fit for the three cantilever configurations.

6.1.3 Zero-crossing method

The natural frequency of the measured waveform is found with the zero-crossing method, which locates the points at which the signal passes through the zero-amplitude line. In a complete vibration cycle, this happens twice: once as the waveform moves from positive to negative and once as it moves back in the opposite direction. The interval between alternating zero crossings gives the vibration period, whose inverse is the natural frequency. To make those crossings easier to detect accurately, the raw signal is first passed through a Butterworth filter that reduces noise. Around each crossing, the algorithm selects two neighbouring data points with timestamps t_1 and t_2 , then uses linear interpolation to determine the corresponding zero-crossing time, t_0 .

$$t_0 = t_1 - \frac{v_1}{m} \quad (17)$$

where:

t_0 is the time at zero-crossing

t_1 is the time corresponding to the immediate next data set.

m is the slope of the two points between the zero-crossing point.

Now, the loop returns an array with all the zero-crossing points and their corresponding timestamp. The time period can be found by taking the difference between each consecutive point and then taking its average to find frequency.

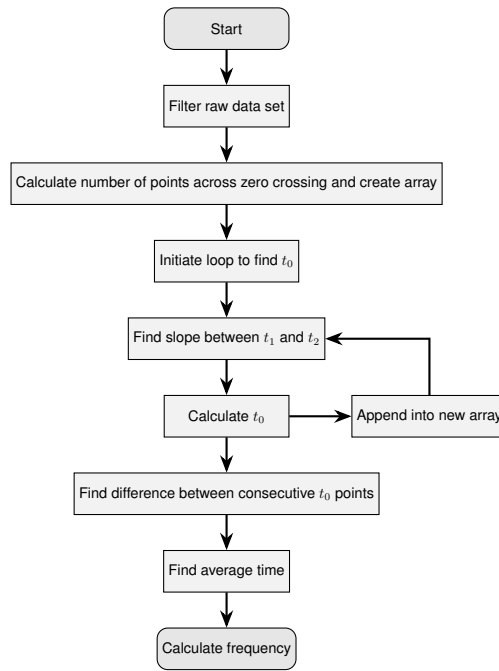


Figure 10: Flowchart for the zero-crossing frequency calculation.

Zero-crossing for the three cantilevers

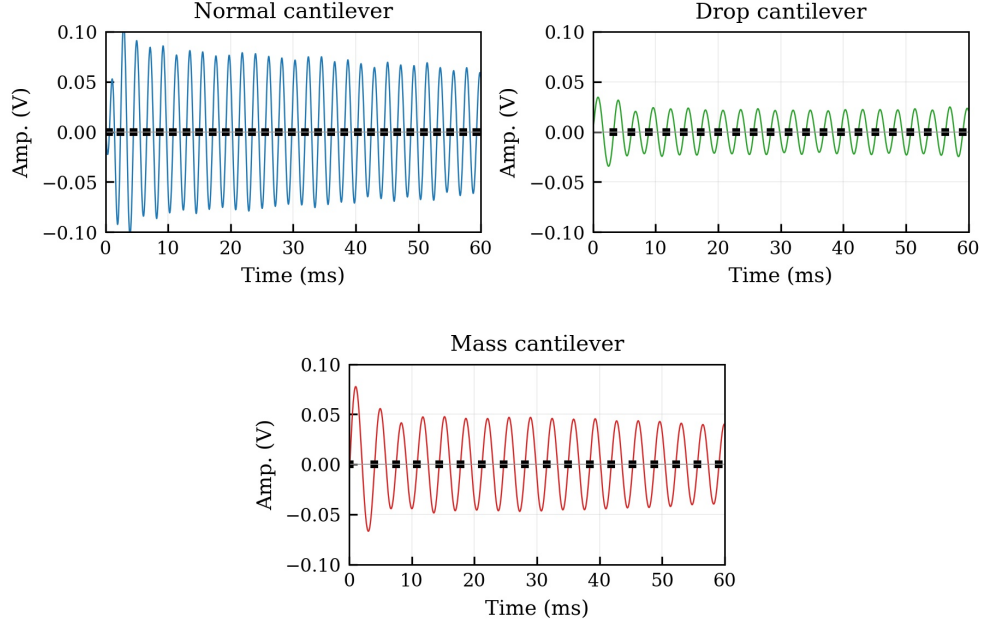


Figure 11: Zero-crossing method applied to the three cantilever configurations.

6.1.4 Parabolic interpolation of the FFT peak

Parabolic fitting algorithm assumes the fact that the peak of the Fourier transform is a gaussian curve. Therefore considering the formula below for a gaussian curve:

$$Y = A \cdot e^{-\frac{(f-f_0)^2}{2\sigma^2}} \quad (18)$$

where f_0 is the true peak frequency. If we take the logarithm on both sides of the equation,

$$\ln Y = \ln A - \frac{(f-f_0)^2}{2\sigma^2} \quad (19)$$

which is the equation of a parabola in the form $ax^2 + bx + c$. We know that the first order derivative is zero when $x = x_{max}$. Therefore, solving the equation for $x = x_{max}$ we get,

$$x = -\frac{b}{2a} \quad (20)$$

Solving for a and b ;

$$x = \delta = \frac{L_{k-1} - L_{k+1}}{2 \cdot (L_{k+1} - 2L_k + L_{k-1})} \quad (21)$$

and the refined peak frequency is then given by,

$$f_{\text{refined}} = (k + \delta) \Delta f. \quad (22)$$

The algorithm first selects the desired timestamp manually for processing after user input, and then takes the Fourier transform first. This array of FFT data points is then looped through to find the peak data point first, say k . The idea is then to find the neighbouring bins say $k+1$ and $k-1$. The next step is then to simply evaluate fractional shift δ using equation (21). If δ is positive, then the refined frequency is to the right of the calculated peak and vice versa.

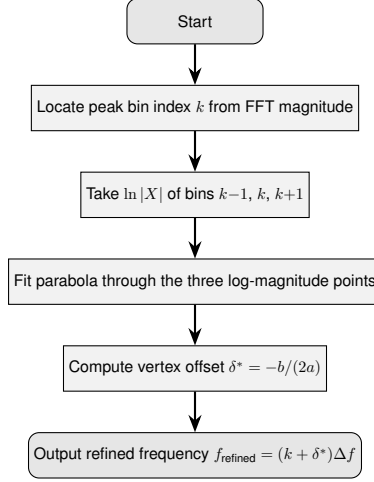


Figure 12: Flowchart for the parabolic interpolation method.

Parabolic interpolation for the three cantilevers

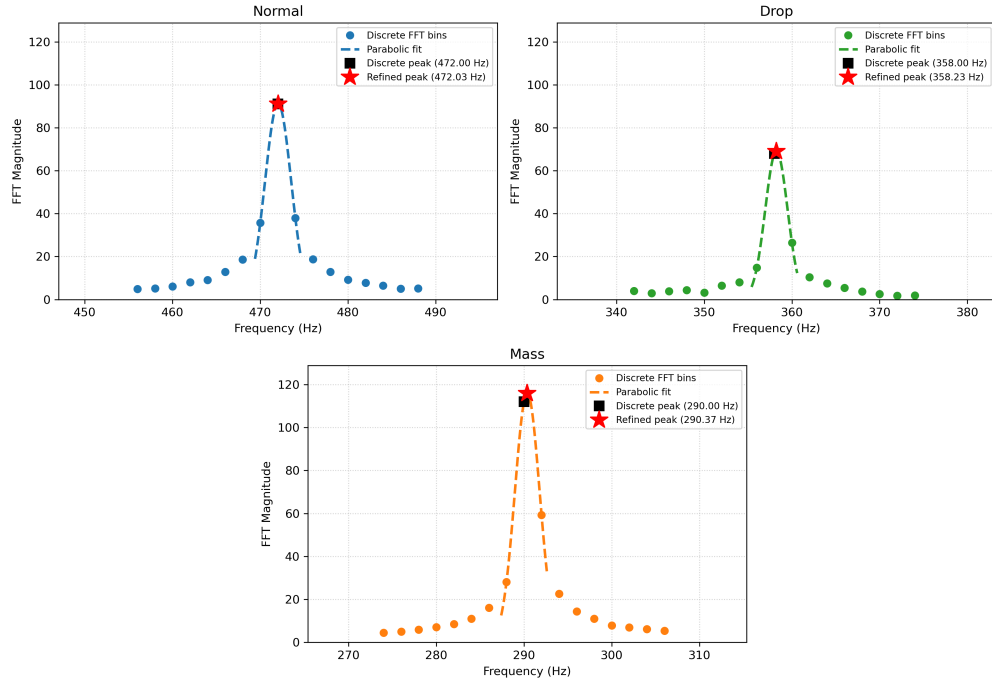


Figure 13: Parabolic interpolation of the FFT peak for the three cantilever configurations, showing the discrete FFT bins, the fitted log-parabolic curve, the standard discrete peak, and the refined sub-bin peak.

Table 4 compares the standard discrete-bin peak frequency with the parabolically refined estimate for the three configurations. The correction is small in all three cases (well under one FFT bin width), which confirms that the record length used was already sufficient to resolve each cantilever's resonance to good accuracy; the refined values are nonetheless adopted as the final frequency estimates because they remove the residual bin-quantisation bias.

Table 4: Standard discrete FFT peak vs. parabolically refined peak.

Configuration	Discrete Peak (Hz)	Refined Peak (Hz)	Correction (Hz)
Normal Cantilever	472.00	472.03	+0.03
Drop Cantilever	358.00	359.23	+0.23
Mass Cantilever	290.00	290.37	+0.37

6.2 Comparison between different cantilevers

The ring-down responses were used to examine the vibration behaviour of the three cantilever configurations, with the results given in Table 5. For each configuration, the natural frequency, maximum amplitude, and damping ratio were determined. At 472.03 Hz, the Normal cantilever had the highest natural frequency; the Mass cantilever had the lowest, at 290.37 Hz. Adding the tip mass increases the system's inertia, so this lower frequency is expected. All three configurations had damping ratios of approximately 0.001 to 0.003, showing that they were lightly damped. The clearest differences were in natural frequency and initial vibration amplitude. Their damping behaviour, by comparison,

Table 5: Comparison of the three cantilever configurations.

Configuration	Natural Frequency (Hz)	Damping Ratio (ζ)	Quality Factor (Q)
Normal Cantilever	472.03	0.0030	168.5275
Drop Cantilever	358.23	0.0021	236.9374
Mass Cantilever	290.37	0.0015	326.8341

7 Conclusion

The experimental analysis of the cantilever clearly indicates the fundamental relationship governing mass and damping characteristics. One notable observation is the decrease in natural frequency with addition of mass at the tip which increased the overall inertia. Similarly, the exact opposite is seen with respect to quality factor (Q).

$$Q = 2\pi \cdot \frac{E_{\text{stored}}}{E_{\text{lost per cycle}}} \quad (23)$$

The observed rise in Q indicates that the energy stored in the cantilever was higher relative to the energy lost due to friction and drag. This rise in stored energy can also be observed in Figure 13 where higher Q -factor cantilever exhibited narrower peak bandwidths. Finally, various signal processing techniques and their limitations was explored and studied.

8 Contributions from team members

Please find below respective contributions from the team members that led to successful completion of the primary objectives.

1. Anveesha Arora: Physical Experimental Setup, Routing and soldering of components and Natural frequency.
2. Donna: Physical experimental setup, data collection for Normal, Drop, Mass cantilever configurations, Mass Cantilever code development for processing and analyzing data(using Fourier Transform, Peak Detection, Parabolic Interpolation)and evaluated natural frequency and quality factor, compared the results of the three different cantilevers.
3. Georgy Baby Valavathu: Transimpedance stage design, Routing and soldering of components, study of algorithm for data analysis.
4. Nigil: Design and Testing of Transimpedance Amplifier and Soldering of components and contribution of Introduction,Objective and Expected outcome of the project in report documentation.
5. Nithish Kumar Baskaran: Design and Testing of Transimpedance stage and Differential stage amplifier circuit. SPICE validation for component selection. Schematic design and Soldering of amplifier circuit.
6. Nandini Mavillapalle: Developed and implemented data analysis algorithms (Fourier Transform, Peak Detection, and Zero-Crossing), performed natural frequency and quality factor analysis and compared different cantilevers, and prepared the results and report documentation.
7. Nikhila: Physical experimental setup and data collection for the Normal, Drop, Mass cantilevers configurations. Code development for processing and analyzing the Normal cantilever data using Fourier Transform (FFT), evaluation of experimental results, contribution to preparation about comparison of the results of the three different cantilevers.

We would also like to take this opportunity to thank Prof. Friedrich Fleischmann and Prof. Ludger Kempen for giving us valuable feedback and assistance throughout the project.

9 Documentation of Tools and AI Usage

9.1 Generative AI Usage

The following table documents the use of GenAI tools during the research and writing process of this report.

Table 6: Documentation of Generative AI Usage

Phase in the work process	Usage / Task Description	GenAI /Research Assistant
Idea generation and brainstorming of Amplifier Design	Construction of the 1D Pi-network equivalent circuit for the S3932 PSD and reasoning of passive component selection in LTspice.	Z - GLM 4.7
Code generation for signal processing and analysis	Algorithm to run python code for various signal processing techniques - Zero crossing, Fourier transform, etc	Gemini Flash 3.6 & Claude Sonnet 5

9.2 Software and Hardware Tools

The following table documents the use of engineering software tools during the prototyping process of this project.

Table 7: Documentation of Engineering Tools and Software

Tool /Software	Phase /Application	Specific Use Case
LTspice 24.0.09	Circuit Simulation	Simulating the S3932 Pi-network, performing AC frequency sweeps (.ac dec), transient step analysis (.tran), and executing parameter sweeps (.step) to determine Phase Margin and optimal C_f values.
KiCad 10.0	Schematic & PCB Design	Drafting the final hardware schematic.
GitHub	Version Control	Maintaining a common codebase for improving & working with python code
Visual Studio Code	Running Jupyter	Writing and running the python code for different signal processing techniques discussed in the report.

References

- [1] David J. Ewins. *Modal Testing: Theory, Practice and Application*. Research Studies Press.
- [2] Jerald G. Graeme. *Photodiode Amplifiers: Op-Amp Solutions*. New York: McGraw-Hill, 1995.
- [3] Paul Horowitz and Winfield Hill. *The Art of Electronics*. 3rd ed. New York: Cambridge University Press, 2015.
- [4] OSI Optoelectronics. *Position Sensing Photodiodes (PSDs) — Characteristics*. Hawthorne, CA.
- [5] Singiresu S. Rao. *Mechanical Vibrations*. Pearson Education.
- [6] “Technical Note / PSD (Position Sensitive Detectors)”. In: (). URL: https://www.hamamatsu.com/content/dam/hamamatsu-photonics/sites/documents/99_SALES_LIBRARY/ssd/psd_kpsd9001e.pdf.
- [7] Texas Instruments. *Stability Analysis of Voltage-Feedback Op Amps*. Dallas, TX: Texas Instruments Incorporated, 2016.
- [8] Paul West. *Introduction to Atomic Force Microscopy: Theory, Practice, Applications — How AFM Works*.